AURA Twin

History + wearable state.
Fresh-patient evidence.
A challengeable prediction.

AURA-SYNTHETIC-2.0.0

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Generated data only · No clinical validation or treatment recommendation

Give changing signals their historical context.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

Historical context

Age, laboratory history and condition duration explain the baseline.

02

Changing observations

Glucose, HR, HRV, activity and sleep show recent state.

03

A reviewable forecast

One patient state supports prediction, context and evidence.

A clinician-oriented proof of concept; workflow benefit has not been measured.

Predict a defined event, at a defined time.

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EXACT-HORIZON EVENT

Glucose ≥180 mg/dL at t + 120 minutes

Not any excursion during the next two hours.

1 **Four forecasts**

+30 / +60 / +90 / +120 minutes

2 **One event probability**

Calibrated estimate for the exact two-hour reading

3 **Explicit convention**

70–180 is common CGM context; inclusive 180 is our label.

Context: NIDDK, Clinical Targets for Continuous Glucose Monitoring Data. Individual targets can differ.

Synthetic data with deliberately varied routines.

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01

Own seeded simulator

Wholly generated people and signals. No real records or device accounts.

02

Six routine families

Regular, delayed, jittered, daily changes/skips, variable sleep/activity.

03

No hidden meal oracle

Future meals and simulator latent state are excluded from inputs.

Final evaluation uses fresh identities within the declared routine mixture.

Only the available past enters the prediction.

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01

Recent dynamics

Lagged CGM, slopes, variability, HR/HRV, steps and sleep.

02

Previous-day observations

Horizon-aligned readings are at t+h–24h.
Observed past, not today’s future.

03

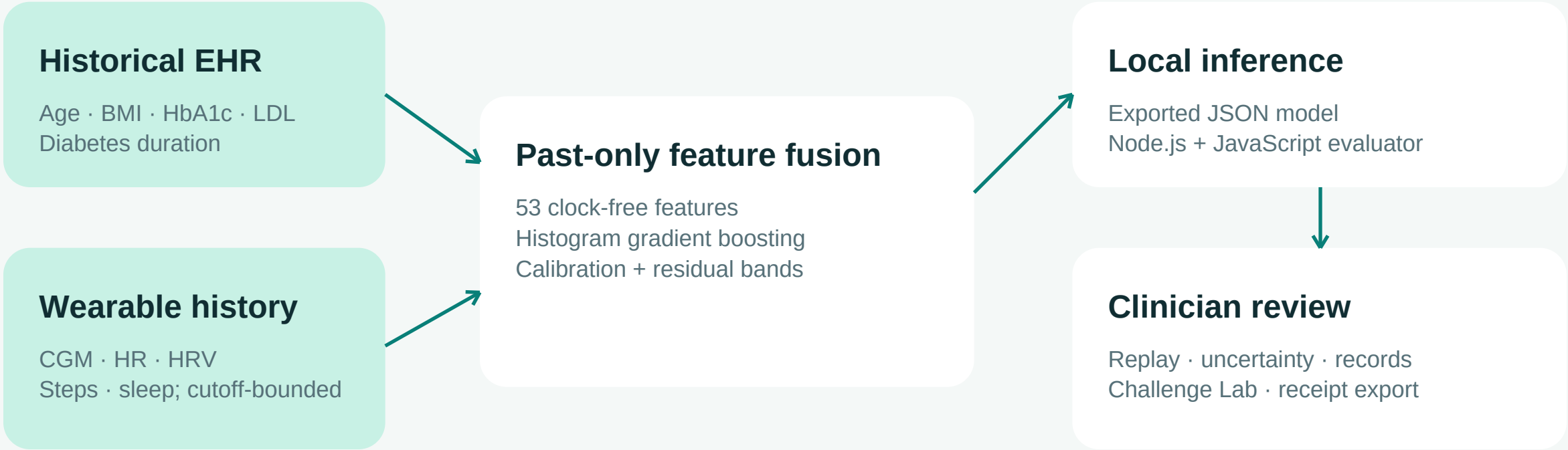
Explicit boundary

Changing future readings must not change a frozen prediction.

Past-only does not mean future behavior becomes predictable.

Two streams. One inspectable inference path.

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No future meals, future readings or hidden simulator state enter inference.

Select and calibrate before opening the final cohort.

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256

Fit

Training identities
Candidate fitting only

48

Select

Separate identities
Choose candidate only

48

Calibrate

Separate identities
Probability, bands, policy

80

Final test

New indices 1000–1079
Generate after model freeze

Every routine variant and window for an identity stays in one partition.

Fresh identities within a declared six-routine mixture; not a test of every unseen behavior.

A clock-free candidate earns the final evaluation.

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01

Remove a shortcut

Clock-free inputs replace explicit time-of-day harmonics.

02

Select on development

Compare candidate models on separate selection identities.

03

Freeze before final

Calibration and policy are fixed before fresh final scoring.

Selection, calibration and final reporting have different patient partitions.

Measure error and the review-policy tradeoff.

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Model	+120 MAE ↓	AUROC ↑	Brier ↓
Current fused	11.62	0.936	0.0436
Sensors only	11.73	0.929	0.0448
EHR only	18.48	0.834	0.0615
Persistence	20.32	0.583	0.1138
Frozen v1	18.30	0.848	0.0737

Policy 0.16: recall 78.7% · precision 40.5%

Units: mg/dL. Repeated windows are correlated. Scores do not establish clinical validity.

Compare models on the same fresh routine cases.

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Routine	Current MAE	Legacy MAE	Persistence	Band cover
Regular	10.06	10.88	21.68	92.28%
Meals +2h	9.72	21.92	21.47	92.84%
Meals +4h	9.29	22.92	19.57	93.34%
Jittered meals	13.97	17.05	21.11	85.31%
Daily changes + skips	13.12	17.64	19.45	86.75%
Variable sleep/activity	13.57	19.39	18.65	86.01%

All rows use the same fresh patient identities. Routine families were declared during development.

Show uncertainty without promising coverage.

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89.42%

Observed coverage

Nominal 90% descriptive residual bands.
Measured on the fresh final cohort.

61.7 mg/dL

Mean band width

At the two-hour horizon.
Broader uncertainty can accompany irregular behavior.

0.16

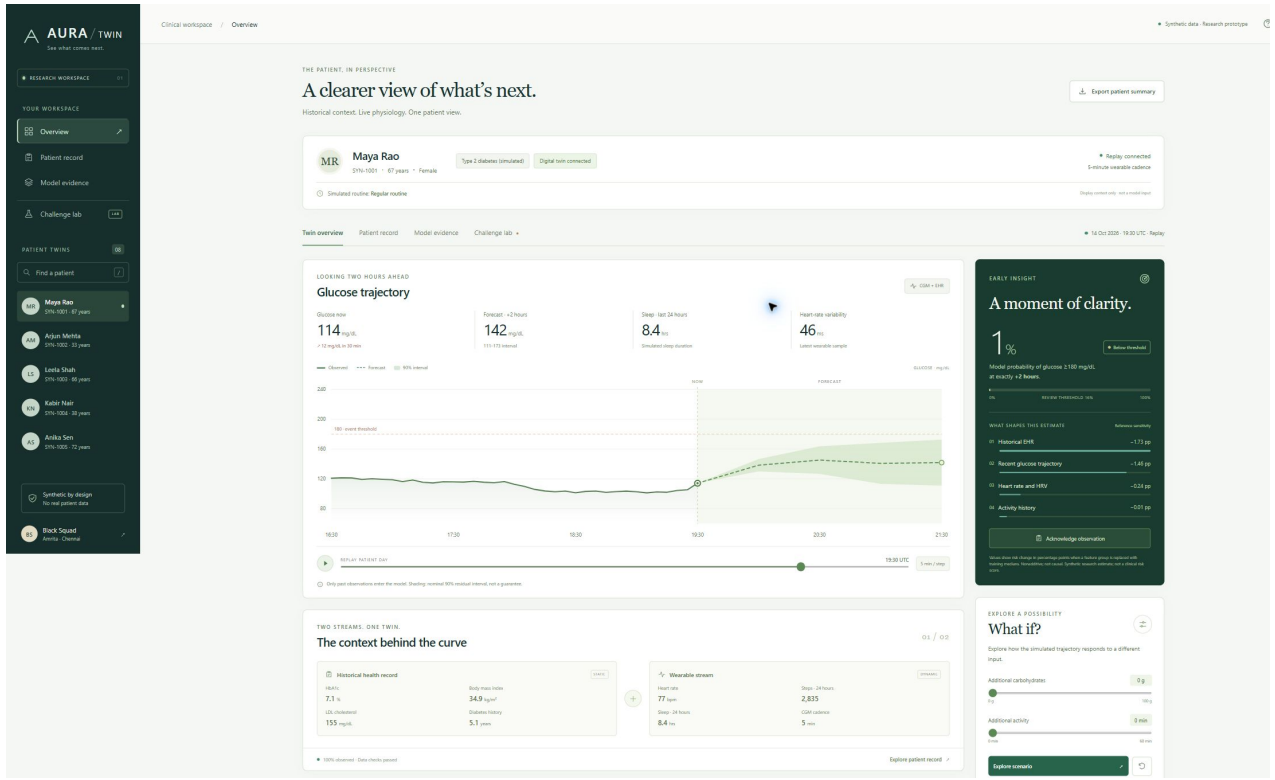
Review policy

Threshold fixed on calibration patients.
Not a validated clinical trigger.

Coverage is marginal. No guarantee per patient, subgroup or shifted population.

An overview designed for a clinical review.

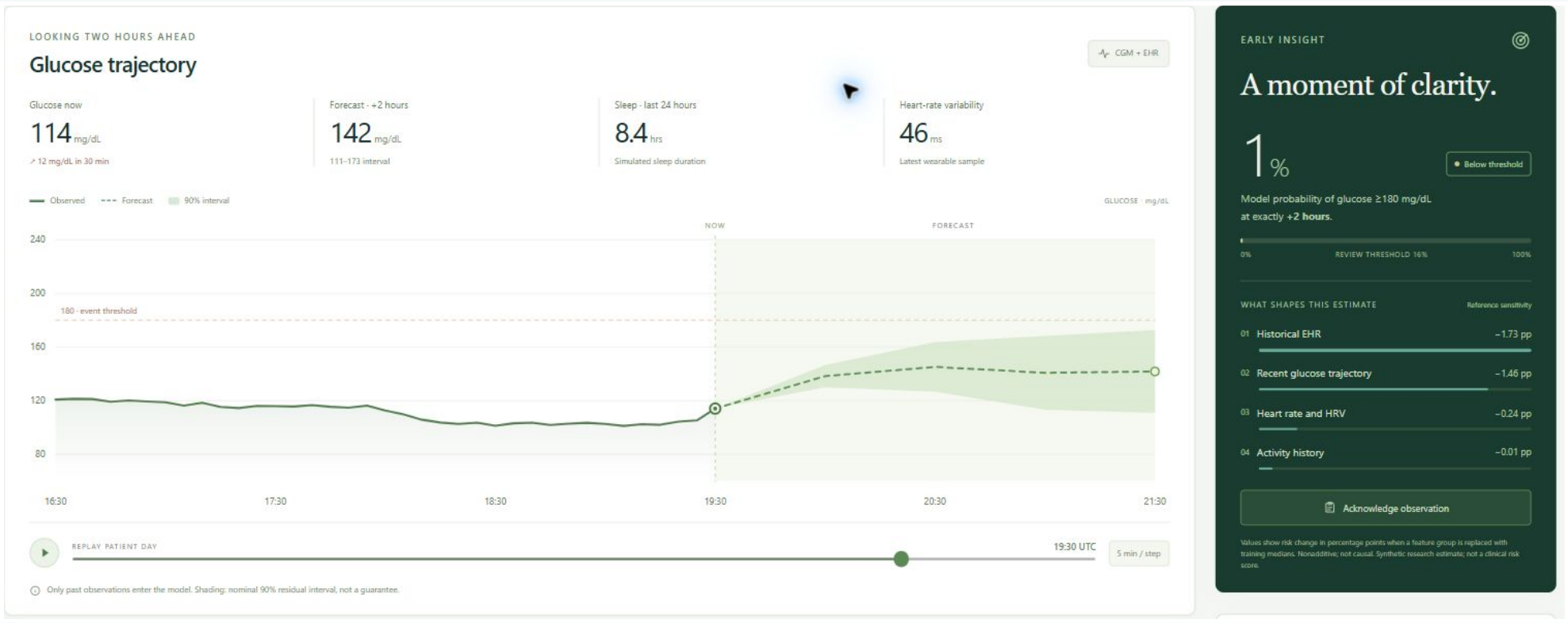
Current model aura-synthetic-2.0.0 · synthetic benchmark evidence



Actual local interface · all observations are generated for this demonstration

Replay makes the information boundary tangible.

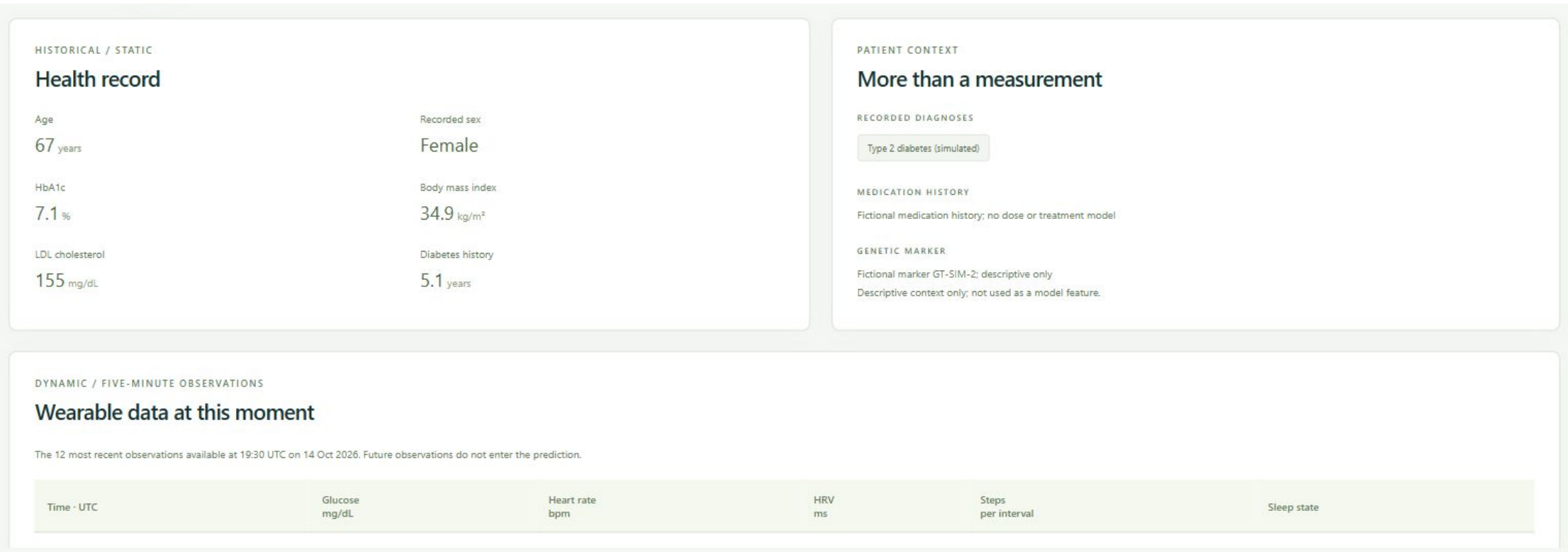
Current model aura-synthetic-2.0.0 · synthetic benchmark evidence



Actual local interface · all observations are generated for this demonstration

Inspect context without turning it into a cause.

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Actual local interface · all observations are generated for this demonstration

Explore an assumption. Do not infer a treatment.

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01

Baseline forecast

The fitted model provides the numerical trajectory and event probability.

Reset restores this baseline.

02

Illustrative sensitivity

Meal/activity sliders apply hand-authored arithmetic assumptions.

Trained risk remains unchanged.

No future meal knowledge. No causal benefit estimate. No treatment or dosing recommendation.

Freeze the prediction. Then reveal the outcome.

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01

Choose

Patient, cutoff, sensor case.
Copy and truncate history.

02

Freeze

Predict before seeing outcomes.
Preserve output + hashes.

03

Reveal

Original recorded future.
No recomputation.

04

Compare

Model vs persistence.
Interval hits + receipt.

Public data: this enforces a computation boundary, not secure study blinding.

Four sensor cases expose different failure modes.

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Clean replay

An honest baseline; a local window may lose to persistence.

10-minute CGM gap

Forward fill permitted; show degraded quality.

30-minute dropout

Stale signal; withhold forecast and probability.

Undetected sensor bias

Plausible +30 mg/dL input shift may pass freshness checks.

The reference future stays unchanged. Persistence sees the same perturbed current value.

Make the experiment reproducible offline.

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01

What can be checked

Raw model bytes → SHA-256
Canonical input prefix → SHA-256
Recomputed forecast and revealed score
Mismatch detection for altered artifacts

02

What is not established

No digital signature
No independent timestamp
No trusted authorship guarantee
No clinical audit certification

```
node scripts/verify_receipt.mjs path/to/receipt.json
```

The evidence travels with the implementation.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

Reproduce the model

Generator, selection protocol, calibration and frozen artifacts.

02

Verify implementation

Python/JavaScript parity, prefix invariance and challenge tests.

03

Run locally

Node.js; zero runtime dependencies and no external model account.

Metrics and fingerprints identify the exact evaluated artifact.

Improved synthetic robustness still has limits.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

Unobserved future behavior

Unexpected meals and abrupt behavior remain difficult to anticipate.

02

Synthetic assumptions

Designed physiology and routine families may favor the model.

03

Real-world evidence gap

No clinical, device, outcome or compliance validation.

Synthetic benchmark improvement is not clinical robustness.

Use the remaining failures to guide validation.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

Broader tests

Independent simulators, held-out routine families and device faults.

02

Independent evaluation

Permitted external data, calibration, subgroups and silent studies.

03

Governed workflows

Clinical ownership, access control, audit and usability validation.

These are future validation steps, not implemented clinical capabilities.

A forecast you can inspect.

A working fusion pipeline.
A measured synthetic improvement.
A reproducible challenge.

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